

BROOKS, AB, Canada

WMO: 714570

Lat: 50.560N

Lon: 111.850W

Elev: 747

StdP: 92.67

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.2	-26.8	-33.9	0.2	-29.6	-30.4	0.2	-26.7	11.9	-4.0	10.7	-4.8	1.8	180	0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	31.7	17.5	29.7	17.0	27.8	16.4	19.6	27.6	18.5	26.5	17.6	25.5	4.1	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.0	13.3	22.4	15.8	12.3	21.1	14.7	11.4	20.2	59.2	27.8	55.6	26.4	52.4	25.6	30.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.1	8.0	DB	-34.5	35.0	4.0	1.8	-37.4	36.2	-39.7	37.3	-41.9	38.3	-44.8	39.5	(4)
(5)			WB	-34.7	22.1	3.9	2.3	-37.5	23.7	-39.8	25.0	-42.0	26.3	-44.8	28.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.9	-9.7	-7.2	-1.7	5.4	11.3	15.6	19.1	18.0	12.8	5.6	-2.4	-8.6
	DBStd	11.84	9.01	7.97	8.09	4.88	4.37	3.24	3.03	3.23	4.34	5.05	7.48	8.34
	HDD10.0	2764	611	482	363	148	37	2	0	1	21	151	374	577
	HDD18.3	5003	869	716	620	387	221	95	27	46	172	394	622	835
	CDD10.0	908	0	0	1	11	77	168	282	248	104	14	2	0
	CDD18.3	106	0	0	0	0	2	12	51	35	6	0	0	0
	CDH23.3	1953	0	0	1	8	108	218	793	630	188	7	0	0
	CDH26.7	647	0	0	0	1	23	50	290	227	56	0	0	0
Wind	WSAvg	3.5	3.3	3.3	3.8	4.1	4.1	3.7	3.4	3.2	3.3	3.5	3.4	3.3
Precipitation	PrecAvg	330	14	11	16	26	43	65	40	37	32	16	17	14
	PrecMax	554	29	23	44	79	90	195	124	86	100	43	45	25
	PrecMin	167	3	1	0	3	7	7	2	1	2	4	3	1
	PrecStd	75	7	5	11	15	23	40	33	19	22	10	11	7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.7	12.1	18.1	24.3	29.5	30.9	34.3	33.8	31.3	24.0	17.5	11.2
		MCWB	4.9	5.6	8.6	11.8	14.7	17.7	19.2	17.3	15.6	12.9	9.3	4.9
	2%	DB	6.9	8.1	14.7	21.0	26.4	28.0	32.0	31.4	28.2	20.2	13.5	7.9
		MCWB	3.4	3.2	7.1	10.2	13.9	16.3	18.4	17.1	14.9	11.1	7.1	3.5
	5%	DB	5.0	5.5	11.9	17.9	23.7	25.8	29.9	29.3	25.3	17.4	10.6	5.5
		MCWB	2.3	2.1	5.5	8.5	12.7	15.4	17.9	16.9	14.1	9.7	5.5	2.2
	10%	DB	3.1	3.4	8.9	15.1	21.0	23.8	27.9	27.1	22.4	14.7	7.8	3.3
		MCWB	1.0	0.8	4.2	7.1	11.5	14.5	17.2	16.5	12.9	8.2	3.7	0.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	5.5	5.8	9.2	12.6	16.7	20.0	22.0	20.3	17.5	13.8	9.6	6.0
		MCDB	8.8	11.1	16.6	22.3	25.1	27.2	29.9	27.8	28.0	21.3	16.2	10.2
	2%	WB	3.7	3.7	7.6	10.7	14.8	17.8	20.1	19.0	15.8	11.9	7.5	3.9
		MCDB	6.4	7.4	13.7	19.6	23.6	25.4	28.4	26.8	25.2	18.9	13.1	7.3
	5%	WB	2.5	2.2	5.9	9.1	13.5	16.6	19.0	18.1	14.6	10.1	5.7	2.3
		MCDB	4.9	5.3	11.2	16.8	22.0	23.7	27.2	26.2	23.3	16.6	10.3	5.4
	10%	WB	1.0	0.9	4.3	7.6	12.2	15.4	18.1	17.1	13.4	8.6	3.9	0.8
		MCDB	3.2	3.3	8.6	14.1	19.9	21.9	26.0	25.1	21.4	14.0	7.4	3.3
Mean Daily Temperature Range	5% DB	MDBR	11.3	12.0	12.4	14.1	14.3	13.2	14.9	15.3	15.1	13.9	11.8	11.2
		MCDBR	12.7	14.3	16.8	20.2	19.2	16.9	18.6	19.9	21.2	18.3	15.1	12.8
		MCWBR	9.6	10.0	10.0	10.4	8.7	7.4	7.4	8.0	9.4	9.9	9.4	9.0
	5% WB	MCDBR	12.3	13.9	16.0	18.6	17.0	15.3	16.5	17.0	19.1	17.6	14.7	12.7
		MCWBR	9.6	10.1	10.0	10.1	8.0	7.5	7.4	7.7	9.0	9.9	9.6	9.3
Clear-Sky Solar Irradiance	taub	0.272	0.283	0.302	0.351	0.374	0.365	0.363	0.385	0.336	0.296	0.268	0.257	
	taud	2.386	2.426	2.436	2.299	2.283	2.364	2.392	2.303	2.424	2.511	2.479	2.395	
	Ebn at Noon	775	865	909	890	881	891	887	846	858	840	783	740	
	Edh at Noon	61	77	91	118	125	117	112	116	92	70	55	52	
All-Sky Solar Radiation	RadAvg	1.19	2.18	3.54	4.89	5.83	6.27	6.81	5.49	3.93	2.41	1.32	0.91	
	RadStd	0.11	0.20	0.25	0.35	0.39	0.38	0.33	0.35	0.34	0.26	0.12	0.08	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47) Regional (81 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page